# Big Data Analytics (BDA)

*In-class SQL Queries — Lecture Attendance Submission*

| Student Name | PRN | Date | Course |
|---|---|---|---|
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## Assumptions

Queries are written for a single denormalized retail table named **orders**, using the column names from the worksheet: `order_status`, `category`, `revenue`, `customer_segment`, `sales_channel`, `payment_method`, `profit`, `customer_rating`, `city`, `product`. If the Hive / Spark table in your notebook has a different name, replace `orders` with that table name. Syntax is Spark SQL / HiveQL compatible (`LIMIT`).

### Task 1

*Show all delivered orders.*

```
SELECT *
FROM orders
WHERE order_status = 'Delivered';
```

### Task 2

*Show Electronics orders with revenue above 50,000.*

```
SELECT *
FROM orders
WHERE category = 'Electronics'
  AND revenue > 50000;
```

### Task 3

*Count orders by customer_segment.*

```
SELECT
    customer_segment,
    COUNT(*) AS order_count
FROM orders
GROUP BY customer_segment;
```

### Task 4

*Calculate total revenue by sales_channel.*

```
SELECT
    sales_channel,
    SUM(revenue) AS total_revenue
FROM orders
GROUP BY sales_channel;
```

### Task 5

*Calculate total profit by payment_method.*

```
SELECT
    payment_method,
    SUM(profit) AS total_profit
FROM orders
GROUP BY payment_method;
```

### Task 6

*Find average customer rating by city.*

```
SELECT
    city,
    AVG(customer_rating) AS avg_customer_rating
FROM orders
GROUP BY city;
```

### Task 7

*Find the top 10 products by total profit.*

```
SELECT
    product,
    SUM(profit) AS total_profit
FROM orders
GROUP BY product
ORDER BY total_profit DESC
LIMIT 10;
```

### Task 8

*Find the top 10 city/category combinations by revenue.*

```
SELECT
    city,
    category,
    SUM(revenue) AS total_revenue
FROM orders
GROUP BY city, category
ORDER BY total_revenue DESC
LIMIT 10;
```

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## Notes

If a column in your notebook differs (for example `product_name` instead of `product`, or `status` instead of `order_status`), change only that identifier.